Two clocks for quantum information: a finite action law for acquisition through mixed collision baths

Research construction note

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Abstract

The rate at which an open quantum system forgets its state does not determine the rate at which its physically accessible mixed environment acquires that state. We resolve this distinction for a finite collision interface. A finite source interacts once with each fresh maximally mixed register; initial purifiers and the final source are unavailable to the receiver. The receiver may retain all emitted correlations, perform arbitrary quantum instruments, and send classical feedback selecting later interactions. If round j has interaction action at most a_j on every branch, the entire acquired channel is close to a replacer when $\sum_j a_j^4$ is small. Accurate recovery requires $\sum_j a_j^4 = \Omega(\log(1/\varepsilon))$ under uniformly weak collisions. For partial SWAP, a mixed dummy and reversed collision circuit give an exact recovered channel with residual $\prod_j (1 - \sin^4 \theta_j)$. Established transpose-recovery inequalities put this receiver within a factor two of every decoder's worst-case channel error. The source forgets on the faster $\prod_j \cos^2 \theta_j$ clock. We specify finite timing, memory, preparation, calibration, and error costs, and separate dense small-system checks from large-resource formula evaluations. The interface is a physical premise; it is not inferred from a reduced generator.

1 Question, outcome, and selection

An arbitrary unknown quantum state is held by a source that repeatedly meets fresh bath registers. A receiver obtains every emitted physical register, with chronological labels and complete coherent access. When can it recover the input after the source itself has become unavailable?

This differs from reversing the whole closed system and from receiving the complete complementary channel of a pure Stinespring dilation. A mixed bath can have an inaccessible purification. Observing source relaxation does not add that purifier to the receiver's resources.

We answer the question for fresh *tracial* registers: each starts maximally mixed. The principal result is a finite action law including all emitted correlations and genuine classical feedback. An exact partial-SWAP construction attains its weak-collision order. Petz recovery and its near-optimality are established machinery; the acquisition bound and the exact physical clock are the substantive calculation.

Throughout, channel error means *half* the diamond distance,

$$d_\diamond(\Lambda, \Gamma) := \tfrac{1}{2} \left\| \Lambda - \Gamma \right\|_\diamond . \tag{1.1}$$

Thus $0 \le d_\diamond \le 1$ for channels. Recovery guarantees cover every reference entangled with the unknown input.

Independent construction before selection. Several different questions were developed before selection. A computational construction connected local unit-rate dynamics to long non-normal transients and succinct prediction hardness; its consequence overlapped established obstructions and required an unprovided persistent-error compiler. A nonquantum construction

proved an exact ambiguity in internal dissipation under feedback through specified force/position ports, even with a known observable and controllable drift. Its completed consequence was an identification obstruction. A separate pure-port weak-export route stopped at a gap between source/reference decoupling and an efficient decoder. The present route supplies both a general limitation and an operation attaining its scale. Companion selection notes retain the independent constructions. Earlier Astra results are premises and comparators, not objects of a new validation campaign.

Supplied access. The model supplies a finite source, fresh independent maximally mixed inputs, known coupling reach, no return path, and unavailable purifiers. The constructive receiver also needs coherent retention. These facts are not inferred from short observations. Section 8 exposes their costs. A strong source SWAP, when available, remains a one-collision export.

2 Interface and universal statement

The source S has fixed finite dimension $d \geq 2$. Round j introduces a finite register E_j in I_{E_j}/e_j, independent of the source, reference, receiver, and other fresh registers. Its purifier is inaccessible. After one collision the receiver obtains E_j; it never returns to the source. The source is discarded after at most N rounds.

The receiver has arbitrary finite quantum memory initialized independently of the input. Between rounds it may apply arbitrary instruments, retain every outcome, and use a *dephased classical history* h to select later collision Hamiltonians and source-only unitary controls. It retains arbitrary coherent output correlations. Retained quantum memory never couples coherently to S, and there is no source measurement/reset or extra source port. Bounded stopping is included by zero-action padding. All spaces and record sets are finite, without a bound on their size.

Set $\hbar = 1$. After removing source-only and fresh-register-only controls in an interaction picture, suppose

$$a_{j,h} := \int_{\text{round } j} \|H_{j,h}(t)\|_\infty \, \mathrm{d}t \leq a_j \quad \text{for every classical history } h. \tag{2.1}$$

The action is integrated over the *entire encounter with one fresh register*. Subdividing that encounter into time slices does not create fresh collisions or reduce its charged action. The history is fixed during each collision. These are deterministic branchwise integrated bounds, not bounds on average action. Let $\mathcal{C}_N$ be the channel to all retained outputs and records, and put

$$\sigma_N := \mathcal{C}_N(I/d), \qquad \mathcal{R}_{\sigma_N}(X) := \mathrm{Tr}(X)\sigma_N. \tag{2.2}$$

Define

$$q_j := \min\{1, 4a_j^4\}, \quad b_N := \prod_j (1 - q_j), \quad \alpha_N := 1 - b_N, \quad c_d := 1 - d^{-2}. \tag{2.3}$$

Theorem 2.1 (Adaptive tracial acquisition law)**.** *The stated interface obeys*

$$d_\diamond(\mathcal{C}_N, \mathcal{R}_{\sigma_N}) \leq \min\left\{1, \tfrac{1}{2}\sqrt{(d^2-1)\alpha_N}\right\}, \tag{2.4}$$

$$\inf_{\mathcal{R}} d_\diamond(\mathcal{R}\mathcal{C}_N, \mathrm{id}) \geq c_d(1 - \sqrt{\alpha_N}) \geq \tfrac{c_d}{2} b_N. \tag{2.5}$$

The infimum includes every CPTP decoder on the entire receiver output, without computational, memory, or ancillary-resource restrictions.

Corollary 2.2 (Action required for accurate recovery)**.** *If $4a_j^4 \leq q_* < 1$ for all j, then*

$$\inf_{\mathcal{R}} d_\diamond(\mathcal{R}\mathcal{C}_N, \mathrm{id}) \geq \frac{c_d}{2} \exp\left[-\frac{4\sum_j a_j^4}{1 - q_*}\right]. \tag{2.6}$$

Achieving $0 < \varepsilon < c_d/2$ requires

$$\sum_j a_j^4 \geq \frac{1-q_*}{4} \log \frac{c_d}{2\varepsilon}. \tag{2.7}$$

Proof. Apply $\log(1-q) \geq -q/(1-q_*)$ to the product in (2.5). □

The constants depend on source dimension, not total bath dimension. The conclusion changes with nontracial fresh inputs, accessible preparation labels or purifiers, returned registers, coherent feedback, or extra source access. Strong collisions are allowed, but $q_j = 1$ makes this bound uninformative. Section 3.6 permits history-dependent budgets.

3 Proof of the adaptive acquisition law

3.1 A contracting weighted operator norm

For a positive baseline ω, on its support define

$$\|X\|_\omega^2 := \operatorname{Tr}(X^\dagger X \omega^{-1}). \tag{3.1}$$

Inverses are support inverses; the weight need not have unit trace.

Lemma 3.1 (Data processing and source projection)**.** *Every CPTP map Φ satisfies*

$$\|\Phi(X)\|_{\Phi(\omega)} \leq \|X\|_\omega . \tag{3.2}$$

Appending a fixed state is an isometry, as is conjugation by a unitary commuting with ω. If $\operatorname{Tr}\omega = 1$, then $\|X\|_1 \leq \|X\|_\omega$. For $\omega = I_S/d \otimes \sigma_M$, the map

$$\mathcal{P}X = I_S/d \otimes \operatorname{Tr}_S X, \qquad \mathcal{Q} = I - \mathcal{P} \tag{3.3}$$

is an orthogonal projection in this norm, and

$$\|\mathcal{P}X\|_\omega^2 = \|\operatorname{Tr}_S X\|_{\sigma_M}^2 . \tag{3.4}$$

Receiver-only maps commute with both projections and contract their norms separately.

Proof. The variational formula

$$\|X\|_\omega^2 = \sup_Y \{2 \operatorname{Re} \operatorname{Tr} X^\dagger Y - \operatorname{Tr} \omega Y^\dagger Y\}$$

and the Schwarz inequality for the unital CP adjoint $\Phi^\dagger$ prove (3.2). Pull an output test Y back and use $\Phi^\dagger(Y^\dagger Y) \geq \Phi^\dagger(Y)^\dagger \Phi^\dagger(Y)$. Restriction to supports or regularization handles singular baselines. The isometries follow by substitution. The trace-norm bound follows from Hilbert–Schmidt Hölder applied to $(X\omega^{-1/2})\omega^{1/2}$. If $\operatorname{Tr}_S Z = 0$, its weighted inner product with $I/d \otimes A$ is zero. This proves orthogonality; substitution proves (3.4). Receiver maps commute with the source trace, proving the last assertion. □

3.2 The reference process stays product

Run the same protocol with source input I/d. At every stage the joint baseline is

$$\omega = I_S/d \otimes \sigma_M. \tag{3.5}$$

In each classical history block, the fresh source/register factor is $I_S/d \otimes I_{E_j}/e_j$, invariant under the selected unitary. Receiver maps change only σ_M, and source-only unitaries preserve I/d. This proves (3.5) inductively. A spectator reference may be included in M, with factor ρ_R in the baseline.

The selected Hamiltonian commutes with the baseline within each classical block. This is the point at which genuine classical control matters. The same statement does not hold for a coherent control register entangled with the source or for a coupling to retained quantum receiver memory.

3.3 One collision exports only second-order amplitude

Lemma 3.2 (Fresh tracial transfer)**.** *For one history block with action a, append I_E/e and let $y = \|\mathcal{P}X\|_\omega^2$, $z = \|\mathcal{Q}X\|_\omega^2$. The collision changes these to*

$$y' = y + t, \qquad z' = z - t, \qquad 0 \le t \le \min\{1, 4a^4\}\, z. \tag{3.6}$$

Proof. The old $\mathcal{P}X \otimes I_E/e$ commutes with the unitary and is unchanged. For every Hamiltonian on SE,

$$\mathcal{P}[H, \mathcal{Q}X \otimes I_E/e] = 0. \tag{3.7}$$

Expand H in source and fresh-register matrices and $\mathcal{Q}X$ in source and old-memory matrices. Each partial source trace is the trace of a commutator, hence zero. Exact traciality and the stated coupling reach are both used here.

Write $\mathcal{L}_t = -i[H(t), \cdot\,]$ and $\mathcal{U}_t = \mathrm{Ad}_{U_t}$. The exact Volterra identity is

$$\mathcal{U}_T = I + \int_0^T \mathcal{L}_s \,\mathrm{d}s + \int_{0\le r\le s\le T} \mathcal{L}_s\mathcal{L}_r\mathcal{U}_r \,\mathrm{d}r\,\mathrm{d}s.$$

The propagators are weighted-norm isometries and $\|\mathcal{L}_t Z\|_\omega \le 2\,\|H(t)\|_\infty\,\|Z\|_\omega$. Apply $\mathcal{P}$ to input $\mathcal{Q}X \otimes I_E/e$. Its zeroth and first terms vanish. The remainder has norm at most $2a^2\,\|\mathcal{Q}X\|_\omega$. Projection and unitarity also bound it by $\|\mathcal{Q}X\|_\omega$, proving the bound on t. The transfer is orthogonal to the old $\mathcal{P}$ component:

$$\langle \mathcal{P}X, \mathcal{P}\,\mathrm{Ad}_U\,\mathcal{Q}X\rangle_\omega = \langle \mathrm{Ad}_{U^{-1}}\,\mathcal{P}X, \mathcal{Q}X\rangle_\omega = 0,$$

with the appended factor understood. Total norm is preserved, giving (3.6). □

3.4 All rounds, receiver processing, and references

Apply Lemma 3.2 separately within each history block, then sum. This order matters: the maximum integrated branch action need not equal the integral of the maximum instantaneous branch norm. Receiver instruments contract the two components separately. If the initial $\mathcal{P}$ component is zero and the initial total squared norm is K, the total never exceeds K, and

$$y_{j+1} \le y_j + q_j(K - y_j). \tag{3.8}$$

Induction gives

$$\|\mathcal{P}X_{\mathrm{final}}\|_\omega^2 \le K\alpha_N. \tag{3.9}$$

No assumption of independence between emitted registers was made.

For input ρ_{SR}, set

$$X_0 = \rho_{SR} - I_S/d \otimes \rho_R, \qquad \omega_0 = I_S/d \otimes \rho_R.$$

Then $\mathcal{P}X_0 = 0$ and

$$\|X_0\|_{\omega_0}^2 \le d^2 - 1. \tag{3.10}$$

For example, purify the input and use its Schmidt decomposition, of rank $r \le d$. Direct substitution gives $dr - 1$ for the centered norm; discarding part of the reference contracts it. Equivalently, source-basis pinching gives $\rho_{SR} \le d^2\omega_0$, which implies $\mathrm{Tr}(\rho_{SR}^2\omega_0^{-1}) - 1 \le d^2 - 1$. These domination relations also ensure support compatibility throughout.

The final source trace has exactly the norm of the $\mathcal{P}$ component. Equations (3.9)–(3.10) and Lemma 3.1 therefore give, for every reference and input,

$$\|(\mathrm{id}_R \otimes \mathcal{C}_N)(\rho_{SR}) - \rho_R \otimes \sigma_N\|_1 \le \sqrt{(d^2-1)\alpha_N}. \tag{3.11}$$

The state characterization of diamond distance for a difference of channels proves (2.4). There is no extra dimension factor from a Choi-to-diamond conversion.

3.5 A bound on every decoder

Let Φ_d be a normalized maximally entangled state on AS, and write

$$\rho_{AB} = (\mathrm{id}_A \otimes \mathcal{C}_N)(\Phi_d), \quad \omega = I_A/d \otimes \sigma_N, \quad p = d^{-2}, \quad c = 1 - p.$$

Equation (3.9) bounds its centered energy by $E := \|\rho_{AB} - \omega\|_\omega^2 \le (d^2 - 1)\alpha_N$. For any CPTP decoder $\mathcal{R}$, define

$$X_{\mathcal{R}} := (\mathrm{id}_A \otimes \mathcal{R}^\dagger)(\Phi_d).$$

Unital complete positivity implies $0 \le X_{\mathcal{R}} \le I$, while $\mathrm{Tr}_A X_{\mathcal{R}} = I_B/d$ gives $\mathrm{Tr}(\omega X_{\mathcal{R}}) = p$. Its entanglement fidelity is $F_{\mathcal{R}} = \mathrm{Tr}\,\rho_{AB} X_{\mathcal{R}}$. Weighted Cauchy–Schwarz, centered at pI, yields

$$F_{\mathcal{R}} - p \le \sqrt{E}\,\sqrt{\mathrm{Tr}\,\omega(X_{\mathcal{R}} - pI)^2} \tag{3.12}$$

$$\le \sqrt{E\,p(1-p)} \le c\sqrt{\alpha_N}. \tag{3.13}$$

The middle step uses $X_{\mathcal{R}}^2 \le X_{\mathcal{R}}$. Testing the decoded Choi state against Φ_d gives $d_\diamond(\mathcal{R}\mathcal{C}_N, \mathrm{id}) \ge 1 - F_{\mathcal{R}}$. Thus (3.13) proves the first inequality in (2.5); $1 - \sqrt{1-b} \ge b/2$ proves the second. This completes Theorem 2.1.

3.6 Pathwise action budgets

The per-round caps can be replaced by a budget that varies along histories. For an actual branch action with $4a_{j,h}^4 < 1$, assign charge

$$c_{j,h} := -\log(1 - 4a_{j,h}^4). \tag{3.14}$$

Theorem 3.3 (Variable action along every history)**.** *If the sum of charges is at most C_* along* every complete history*, including early stopping, Theorem 2.1 holds with*

$$b_N = e^{-C_*}, \qquad \alpha_N = 1 - e^{-C_*}. \tag{3.15}$$

If every action is at most $a_{\max}$, $q_* = 4a_{\max}^4 < 1$, and every history obeys $\sum_j a_{j,h}^4 \le A_*$, it holds with $b_N = \exp[-4A_*/(1-q_*)]$.

Proof. Keep an inert copy of each already dephased history before any erasure or merging of records. Ignoring that archive recovers the actual protocol, so a bound for the enlarged receiver suffices. At a history node h, put $R_h = C_* -$ charge spent, and let y_h, z_h be its unnormalized weighted squared projection norms. Consider

$$V = \sum_h [y_h + (1 - e^{-R_h}) z_h].$$

An instrument splits a node into children with the same remaining budget. Direct-sum data processing contracts y and z separately, so V cannot increase. A collision of charge c changes (y, z, R) to $(y + t, z - t, R - c)$, with $t \le (1 - e^{-c})z$. Its potential change is

$$e^{-R}[e^c t - (e^c - 1)z] \le 0.$$

Initially $V = (1 - e^{-C_*})K$; finally $V \ge \sum_h y_h$. This proves the energy bound and hence the theorem. The quartic-budget version uses $-\log(1-q) \le q/(1-q_*)$. □

This is a worst-case path budget. No expected-action theorem is asserted: quadratic information weights need not track ordinary branch probabilities.

4 Recovery machinery and its precise normalization

This section applies to any finite channel $\mathcal{C}$. It states established transpose-recovery inequalities [3, 4] and supplies proofs to fix normalization. These inequalities are not new claims. Set $\sigma = \mathcal{C}(I/d)$ and $\rho = (\mathrm{id}_A \otimes \mathcal{C})(\Phi_d)$. On the support of σ, the Petz map is

$$\mathcal{D}(Y) = d^{-1}\mathcal{C}^\dagger(\sigma^{-1/2} Y \sigma^{-1/2}). \tag{4.1}$$

It is CP, has $\operatorname{Tr}\mathcal{D}(Y) = \operatorname{Tr}(\Pi_\sigma Y)$, and satisfies $\mathcal{D}(\sigma) = I/d$. Map the unused block to a fixed state to complete it trace preservingly. All outputs of $\mathcal{C}$ lie in its supported block.

Its entanglement fidelity has the exact expression

$$F_{\mathrm{P}} := F_e(\mathcal{D}\mathcal{C}, I/d) = d^{-1} \operatorname{Tr}\Big[\rho(I \otimes \sigma^{-1/2})\rho(I \otimes \sigma^{-1/2})\Big]. \tag{4.2}$$

For Hermitian Z, the arithmetic–geometric mean inequality in the eigenbasis of σ gives

$$\operatorname{Tr}[Z(I \otimes \sigma^{-1/2})Z(I \otimes \sigma^{-1/2})] \le \operatorname{Tr}[Z^2(I \otimes \sigma^{-1})]. \tag{4.3}$$

Indeed each matrix coefficient is multiplied by $(\lambda_i\lambda_j)^{-1/2} \le (\lambda_i^{-1} + \lambda_j^{-1})/2$. Writing $\omega = I/d \otimes \sigma$ and $E = \|\rho - \omega\|_\omega^2$, expansion gives $d \operatorname{Tr}[\rho^2(I \otimes \sigma^{-1})] = 1 + E$. Consequently,

$$F_{\mathrm{P}} \le \frac{1+E}{d^2}. \tag{4.4}$$

For the collision interface this is $F_{\mathrm{P}} \le 1 - c_d b_N$.

Lemma 4.1 (Established sharp transpose bound)**.** *For $F_{\mathrm{opt}} = \sup_{\mathcal{R}} F_e(\mathcal{R}\mathcal{C}, I/d)$,*

$$F_{\mathrm{P}} \ge F_{\mathrm{opt}}^2 + \frac{(1 - F_{\mathrm{opt}})^2}{d^2 - 1}. \tag{4.5}$$

In particular $F_{\mathrm{opt}}^2 \le F_{\mathrm{P}}$. The latter is the Barnum–Knill bound; (4.5) *is Renes's Eq. (13) [3, 4].*

Proof. For a decoder use $X = (\mathrm{id} \otimes \mathcal{R}^\dagger)(\Phi_d)$, so $0 \le X \le I$ and $\operatorname{Tr}_A X = I/d$. Put

$$A = (I \otimes \sigma^{-1/4})\rho(I \otimes \sigma^{-1/4}), \quad B = (I \otimes \sigma^{1/4})X(I \otimes \sigma^{1/4}).$$

Then $\operatorname{Tr} A^2 = dF_{\mathrm{P}}$, $\operatorname{Tr} AB = F_{\mathcal{R}}$, and, with $S = I \otimes \sigma^{1/2}$,

$$\operatorname{Tr} B^2 = \operatorname{Tr} XSXS \le \operatorname{Tr} XS^2 = 1/d.$$

Cauchy–Schwarz gives $F_{\mathcal{R}}^2 \le F_{\mathrm{P}}$. For the sharper inequality set $A_0 = I_A/d \otimes \sigma^{1/2}$. Direct calculation gives

$$\operatorname{Tr} A_0^2 = \operatorname{Tr} AA_0 = 1/d, \quad \operatorname{Tr} A_0 B = d^{-2}, \quad \|A - A_0\|_2^2 = d(F_{\mathrm{P}} - d^{-2}),$$

and $\|B - A_0/d\|_2^2 \le (1 - d^{-2})/d$. Centering Cauchy–Schwarz therefore yields

$$|F_{\mathcal{R}} - d^{-2}| \le \sqrt{(1 - d^{-2})(F_{\mathrm{P}} - d^{-2})}.$$

Rearrangement gives (4.5), including the optimal decoder. □

Combining (4.4) and (4.5) reproduces the sharp bound (3.13). The direct proof in Section 3.5 avoids even this detour. Both calculations show that the relevant new input to recovery theory is the acquired channel's bounded information energy, not a source relaxation rate.

5 A finite decoder for the emitted partial-SWAP bath

Now each fresh register has dimension d, and the known source collision is

$$U_j = \cos\theta_j I + i \sin\theta_j W_{S,E_j}, \tag{5.1}$$

where W is SWAP. Arbitrary known source-only unitaries may occur between collisions. Let U be the complete circuit and $e = d^N$. The receiver retains every emitted register coherently. Its acquired channel is

$$\mathcal{C}(X) = \mathrm{Tr}_S[U(X \otimes I_E/e)U^\dagger], \qquad \mathcal{C}(I/d) = I_E/e. \tag{5.2}$$

This includes the full e-dimensional output, not a product approximation.

The receiver prepares an independent maximally mixed dummy S', applies the inverse circuit in reverse chronological order, and discards the bath:

$$\mathcal{D}(Y) = \mathrm{Tr}_E[U^\dagger(I_S/d \otimes Y)U] = (e/d)\mathcal{C}^\dagger(Y). \tag{5.3}$$

It uses neither the actual source nor its initial bath purifiers, and has no postselection. Known source rotations are reversed on the dummy. Equation (5.3) is exactly (4.1). Such inverse-unitary implementations under product-preserving reference states are established physical recovery machinery [5, 6].

Theorem 5.1 (Exact recovered-channel identity)**.** *Define*

$$r_N := \prod_{j=1}^{N}(1 - \sin^4\theta_j), \qquad \alpha_N := 1 - r_N. \tag{5.4}$$

The receiver in (5.3) *obeys*

$$\mathcal{D}\mathcal{C} = (1 - r_N)\,\mathrm{id} + r_N \mathcal{R}_{I/d}. \tag{5.5}$$

This exact channel identity permits arbitrary known source-only rotations and tensoring with any reference. Moreover,

$$c_d(1 - \sqrt{1 - r_N}) \le \varepsilon_{\mathrm{opt}} := \inf_{\mathcal{R}} d_\diamond(\mathcal{R}\mathcal{C}, \mathrm{id}) \le d_\diamond(\mathcal{D}\mathcal{C}, \mathrm{id}) = c_d r_N. \tag{5.6}$$

Thus $c_d r_N/2 \le \varepsilon_{\mathrm{opt}} \le c_d r_N$.

Proof. On a D-dimensional operator space use normalized Hilbert–Schmidt inner product $\mathrm{Tr}(X^\dagger Y)/D$. The map $J(X) = U(X \otimes I_E)U^\dagger$ is an isometry, with adjoint $J^*(Z) = e^{-1}\,\mathrm{Tr}_E[U^\dagger Z U]$. For $\mathcal{P}Z = I_S/d \otimes \mathrm{Tr}_S Z$, substitution gives

$$J^*\mathcal{P}J = \mathcal{D}\mathcal{C}. \tag{5.7}$$

At any stage, decompose the full source/old-output operator into its $\mathcal{P}$ and $\mathcal{Q}$ parts. The old $\mathcal{P}$ part is unchanged after appending a fresh identity and colliding. Expand the other part as $\sum_a X_a \otimes A_a$, with X_a source traceless and A_a arbitrary on *all* earlier outputs. With $s_j = \sin\theta_j, c_j = \cos\theta_j$,

$$U_j(X_a \otimes I)U_j^\dagger = c_j^2 X_a \otimes I + s_j^2 I \otimes X_a + i c_j s_j [W, X_a \otimes I].$$

The first and commutator terms have zero source partial trace. Hence

$$\mathcal{P}\,\mathrm{Ad}_{U_j}(X_a \otimes I) = s_j^2 I_S \otimes X_a. \tag{5.8}$$

Relabeling S as E_j is an isometry of normalized operator spaces, including arbitrary old-output coefficients. The new term is traceless on E_j, hence orthogonal to the old exported term, which

is identity there. Thus the hidden squared norm is multiplied exactly by $1 - s_j^4$. Source-only rotations preserve the decomposition and norms.

For initially traceless X, induction gives $\|\mathcal{P}J(X)\|_2^2 = (1 - r_N)\,\|X\|_2^2$, with normalized norms. Polarization gives $J^*\mathcal{P}J = (1 - r_N)I$ on the traceless subspace. The identity is fixed and orthogonal to it, proving (5.5).

The recovered channel is a Weyl mixture with identity probability $1 - c_d r_N$, and probability r_N/d^2 for every nonidentity Weyl. A maximally entangled input yields an orthogonal Bell mixture with half trace distance $c_d r_N$ from the original Bell state. This matches the mixture upper bound on diamond distance, proving the exact error and $F_{\mathrm{P}} = 1 - c_d r_N$. Lemma 4.1 gives the left side of (5.6); $1 - \sqrt{1 - r} \geq r/2$ finishes the proof. □

Remark 5.2 (Endpoints and execution). The identity allows all angles. A full SWAP gives $r_N = 0$ and perfect export in one collision; zero interaction gives $r_N = 1$ and optimal error c_d. Logarithmic count formulas assume $0 < |\theta| < \pi/2$. The decoder consumes one input once, with N inverse collision gates and one mixed dummy; it has no payload copies or retries. It needs the actual chronology and inverse circuit. Knowledge of the reduced channel alone does not provide that circuit. Direct register access, routing, and memory are paid separately. No generic efficient Petz implementation is asserted.

6 Independent modules and collective final decoding

Suppose m emitted channels factorize as $\mathcal{C} = \bigotimes_{i=1}^{m} \mathcal{C}_i$, with source dimensions d_i. Inputs can be entangled across modules and with a reference. Each module may use local classical feedback and memory. Active cross-module control is excluded unless the emitted channel remains a product. Final decoding may be an arbitrary joint operation.

Let $c_i = 1 - d_i^{-2}$, and let b_i be any valid residual bound from Theorem 2.1 or 3.3. The reference output is a product, so the Petz map and its entanglement fidelity factor:

$$F_{\mathrm{P}}(\mathcal{C}) = \prod_i F_{\mathrm{P},i} \leq V := \prod_i (1 - c_i b_i). \tag{6.1}$$

Every collective decoder therefore obeys

$$\varepsilon_{\mathrm{opt}}(\mathcal{C}) \geq 1 - \sqrt{V}, \tag{6.2}$$

$$\varepsilon_{\mathrm{opt}}(\mathcal{C}) \geq 1 - D^{-2} - \sqrt{(1 - D^{-2})(V - D^{-2})}, \quad D = \prod_i d_i. \tag{6.3}$$

Here $V \geq D^{-2}$. These follow from Lemma 4.1; they are joint-input channel bounds, not sums of presumed individual decoder errors.

For target error $\varepsilon < 1$, (6.2) and $V \leq \exp(-\sum_i c_i b_i)$ require

$$\sum_i c_i b_i \leq \eta_\varepsilon, \qquad \eta_\varepsilon := -2\log(1 - \varepsilon). \tag{6.4}$$

For equal dimension d, pathwise budgets A_i, and $\kappa = 4/(1 - q_*)$, choose $b_i = e^{-\kappa A_i}$. Jensen's inequality gives the register-scale necessary cost

$$\sum_i A_i \geq \frac{m}{\kappa} \max\left\{0, \log \frac{c_d m}{\eta_\varepsilon}\right\}. \tag{6.5}$$

Since $\eta_\varepsilon = 2\varepsilon + O(\varepsilon^2)$, this is $\Omega(m \log(m/\varepsilon))$ total quartic action. The product-channel premise is essential; cross-module feedback can invalidate (6.1).

Corollary 6.1 (Exact product receiver and optimal clock)**.** *For independent partial-SWAP modules with residuals r_i, define $V_{\mathrm{ex}} = \prod_i (1 - c_i r_i)$. Then*

$$1 - \sqrt{V_{\mathrm{ex}}} \le \varepsilon_{\mathrm{opt}} \le d_\diamond\left((\bigotimes_i \mathcal{D}_i)(\bigotimes_i \mathcal{C}_i), \mathrm{id}\right) = 1 - V_{\mathrm{ex}}. \tag{6.6}$$

The explicit receiver is within a factor $1 + \sqrt{V_{\mathrm{ex}}} \le 2$ of the optimum, even allowing collective final decoding.

Proof. The recovered channel is a product Weyl mixture with identity probability V_{ex}. A product maximally entangled input gives orthogonal Bell errors and exact half-diamond error $1 - V_{\mathrm{ex}}$. This receiver is Petz, so (6.2) supplies the converse. The ratio is $1 + \sqrt{V_{\mathrm{ex}}}$. □

For equal modules and constant $0 < |\theta| < \pi/2$, put $\ell_\theta = -\log(1 - \sin^4\theta)$, so $r = e^{-N\ell_\theta}$. The exact sufficient and necessary thresholds from (6.6) are

$$r \le \frac{1 - (1-\varepsilon)^{1/m}}{c_d}, \qquad \text{sufficient,} \tag{6.7}$$

$$r \le \frac{1 - (1-\varepsilon)^{2/m}}{c_d}, \qquad \text{necessary.} \tag{6.8}$$

Take logarithms, divide by ℓ_θ, round counts up, and replace negative bounds by zero. Since $\ell_\theta = \theta^4 + O(\theta^6)$,

$$N = \Theta\Big(\sin^{-4}\theta \log(m/\varepsilon)\Big) \quad \text{per module in the weak-collision regime.} \tag{6.9}$$

This recovers the full arbitrary input register. It does not permit a smaller payload encoded at the source across modules, and is not a quantum-capacity theorem.

7 Identical source dynamics, different acquisition clocks

7.1 Second-order forgetting and fourth-order recovery

The reduced source map for one partial SWAP with I/d is

$$X \longmapsto \cos^2\theta\, X + \sin^2\theta\ \mathrm{Tr}(X) I/d.$$

After N collisions its visibility is $\beta_N = \prod_j \cos^2\theta_j$, up to the known net source rotation. Its exact distance from the maximally mixed replacer is

$$d_\diamond(\mathcal{S}_N, \mathcal{R}_{I/d}) = c_d \beta_N. \tag{7.1}$$

For constant weak θ, forgetting takes $\Theta(\sin^{-2}\theta \log(1/\varepsilon))$ collisions, while output recovery takes $\Theta(\sin^{-4}\theta \log(1/\varepsilon))$. These operational questions are answered in the same finite physical model.

7.2 An exact source semigroup does not fix the output clock

Set the source-only rotations to identity. Choose collision duration Δ and rate $\gamma > 0$ by

$$\cos^2\theta = e^{-\gamma\Delta}. \tag{7.2}$$

Here $T = N\Delta$ is the ideal source collision clock, the sum of encounter durations. Additional routing, settling, and decoding time is charged in Section 8; it is not included in this ideal clock.

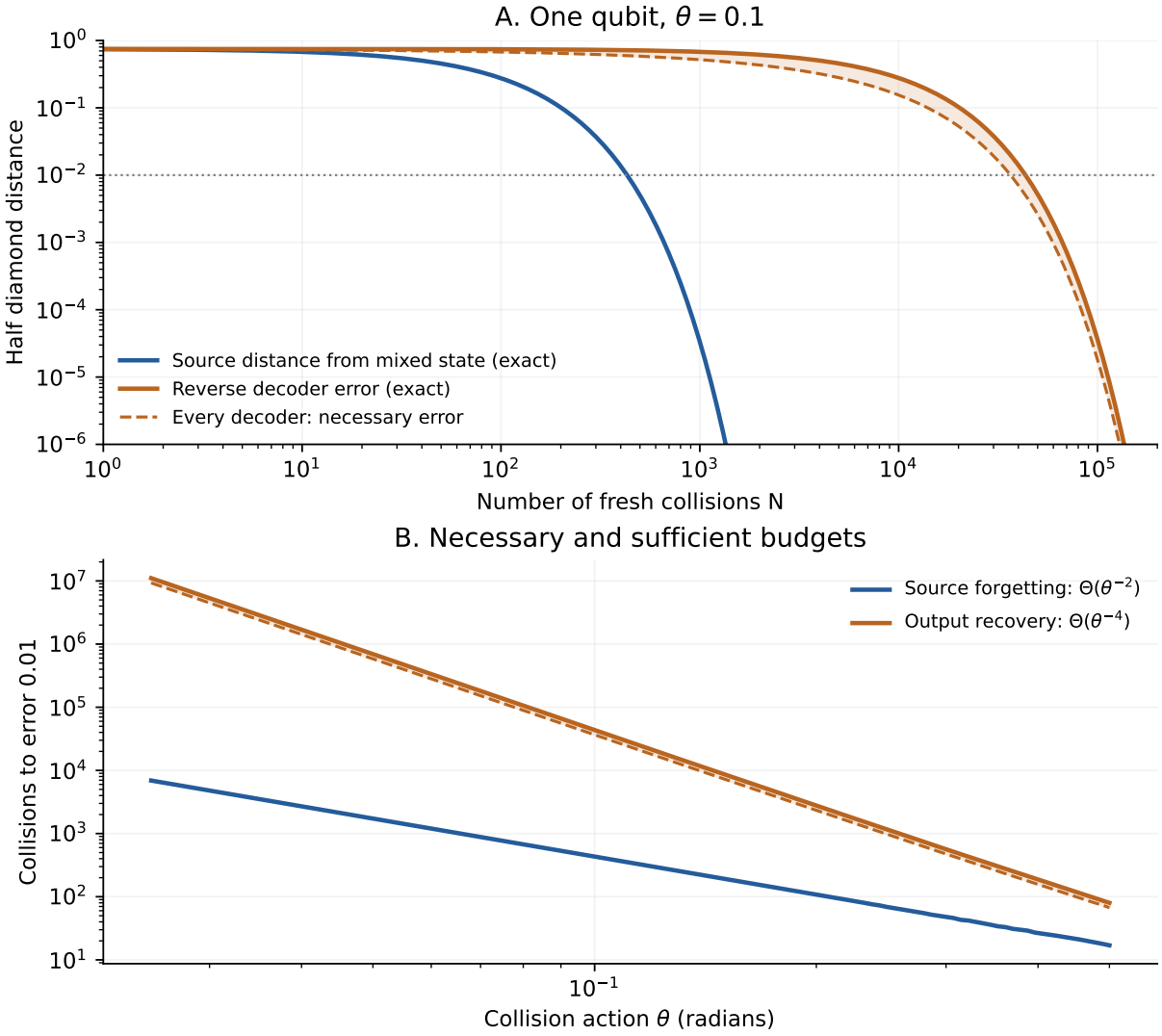


Figure 1: The two clocks, evaluated from exact formulas. Panel A: for one qubit and $\theta = 0.1$, source-to-replacer distance (7.1), explicit recovery error $3r_N/4$, and the lower bound $3(1 - \sqrt{1 - r_N})/4$ for every decoder. Panel B: necessary and sufficient collision counts to reach half-diamond error 0.01. These curves are formula evaluations, not simulations of the displayed many-register systems.

At *every* grid time $T = N\Delta$, the source channel is exactly depolarizing with visibility $e^{-\gamma T}$. The bath residual is instead

$$r_N = \left[1 - (1 - e^{-\gamma\Delta})^2\right]^{T/\Delta}, \qquad \alpha_N = \gamma^2 T\Delta + O(\Delta^2) \tag{7.3}$$

at fixed γ, T. The exact projection recursion gives

$$d_\diamond(\mathcal{C}_N, \mathcal{R}_{I_E/e}) \leq \min\{1, \tfrac{1}{2}\sqrt{(d^2-1)\alpha_N}\}. \tag{7.4}$$

Use (3.11) with exact transfer $\sin^4\theta_j$ instead of $4a_j^4$. The entire bath channel, including every multiregister correlation, approaches a replacer as $\Delta \to 0$ at fixed T, while the source has a nontrivial relaxation law.

This is already exact at a coarse time h. Compare M fresh weak collisions with $\cos^2\theta_M = e^{-\gamma h/M}$, against one collision with $\cos^2\theta_1 = e^{-\gamma h}$. Their source channels at h are identical. The weak protocol has $\alpha_M \sim \gamma^2 h^2/M$ and approaches an input-independent bath. The single collision has $\alpha_1 = (1 - e^{-\gamma h})^2$ and receiver error $c_d[1 - (1 - e^{-\gamma h})^2]$, small for large γh. Repeating blocks preserves equal source channels at every coarse time.

7.3 Simultaneous limits and their physical price

Both source and entire bath marginals can become input independent in one family: take $\Delta \to 0$ and $T = \Delta\lceil\log[1/(\gamma\Delta)]/(\gamma\Delta)\rceil$. Then $\beta_N = O(\gamma\Delta)$ and $\alpha_N = O(\gamma\Delta\log[1/(\gamma\Delta)])$. At every finite size, the joint source-plus-bath inverse restores the input exactly. Input dependence can remain in their joint correlations when neither marginal reveals it. No-hiding concerns the complete pure dilation, including the initially inaccessible purifiers [7]; there is no contradiction.

Holding γ fixed requires $\theta \sim \sqrt{\gamma\Delta}$. A square exchange pulse has strength $\theta/\Delta \sim \sqrt{\gamma/\Delta}$, which diverges. The acquisition exponent per physical time is asymptotically $\gamma^2\Delta$, so collection requires order $\log(1/\varepsilon)/(\gamma^2\Delta)$ time. At fixed bounded strength J, with $\theta = J\Delta$, source forgetting takes order $1/(J^2\Delta)$ and recovery order $1/(J^4\Delta^3)$, up to accuracy logarithms. Repeated-interaction limits and their Hamiltonian scalings are established [8]; the limit does not supply unlimited coupling, switching, or memory.

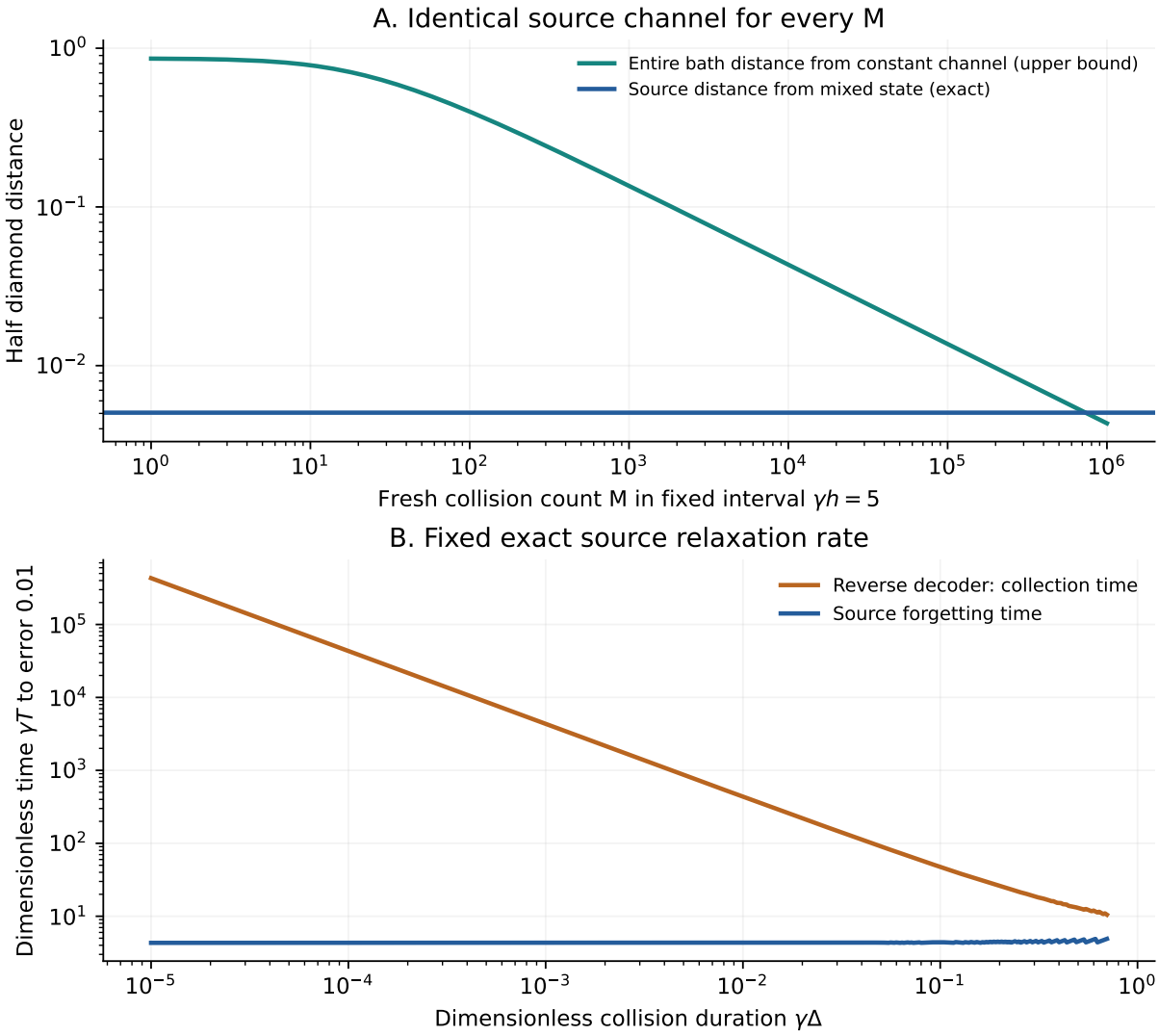


Figure 2: Output access is not determined by the source channel. Panel A: M fresh collisions over $\gamma h = 5$ have the same exact source channel for every M, while the entire bath approaches a replacer. Panel B: fixing the exact grid-time relaxation rate γ, source forgetting has bounded dimensionless time but native output collection time diverges as the collision interval decreases. All quantities use half-diamond error and exact formulas. Fixed-rate shrinking intervals require diverging coupling strength. Native times exclude the additional routing and decoder costs of the finite local implementation.

8 Finite construction and uncertainty budget

8.1 Finite exchange hardware and a routing schedule

This qubit construction is conditional on stated control and error capabilities. It uses $N+2$ cells: source, N memory cells, and a distinct mixed dummy. Initialize and calibrate before loading the unknown payload; otherwise charge its extra waiting time. The nearest-neighbor primitive is

$$U(\phi) = e^{-i\phi W}, \quad H(t) = \hbar g(t) W, \quad \phi = \int g(t)\,\mathrm{d}t, \quad |g| \le J.$$

For spins $W = (I + XX + YY + ZZ)/2$, isotropic exchange up to a scalar. An XY interaction alone is not this primitive. Retain the scalar's constant if the norm budget includes it. Known free evolution is tracked; unknown drift is an error.

The source service supplies one encounter of maximum duration τ_{enc} per fresh logical register, with $|g| \le J_S$. It therefore enforces $a \le J_S \tau_{\mathrm{enc}}$; the encounter cannot be paused, extended, or repeated on the same token. This timing/reach contract is a supplied source-access restriction, not a consequence of bounded strength alone. If duration can reach $\pi/(2J_S)$, a single source SWAP exports the input and is the stronger operation. Receiver routing SWAPs do not supply that source access. Signed receiver exchange implements the inverse in time θ/J_M. Positive-only exchange uses angle $\pi - \theta$, since $U(\pi - \theta) = -U(\theta)^\dagger$, at the longer duration.

Place S at $(-1, 0)$, the memory at $(0,0), (1,0), \ldots, (N-1,0)$, and the dummy at $(0,1)$. The junction has $N+2$ vertices, $N+1$ edges, and maximum degree three. In capture round j, a fresh logical register at cell 1 meets the source. Bubble the emitted token right through the remaining fresh tokens with $N - j$ full SWAPs, stopping before old outputs. Full SWAPs permute factors, so future inputs stay fresh. The final order is $E_N, \ldots, E_1$. Close the source edge permanently.

Inverse-collide the side-branch dummy with E_N, then SWAP the dummy token into cell 1. Inverse-collide with E_{N-1}, move right, and continue; omit movement after the last inverse collision. This implements the reverse logical circuit without source access, measurements, or postselection. All correlations survive until coherent use. The exact routing and total two-qubit gate counts are

$$R_c = N(N-1)/2, \qquad R_d = N - 1, \tag{8.1}$$

$$R = (N-1)(N+2)/2, \qquad G = 2N + R = (N^2 + 5N - 2)/2. \tag{8.2}$$

A strict line $S, E_1, \ldots, E_N, D$ first moves the dummy left through the outputs, adding N routing SWAPs. Let $\tau_c = \theta/J_S$, $\tau_d = \theta/J_M$, and $\tau_w = \pi/(2J_M)$, replacing τ_d by $(\pi - \theta)/J_M$ for positive-only receiver exchange. If τ_{ctl} bounds switching, addressing, and settling per gate, the serial schedule has

$$T_c = N\tau_c + R_c\tau_w + (N + R_c)\tau_{\mathrm{ctl}}, \tag{8.3}$$

$$T_d = N\tau_d + R_d\tau_w + (N + R_d)\tau_{\mathrm{ctl}}, \tag{8.4}$$

$$T_{\mathrm{op}} = T_c + T_d, \qquad V_M \le (N+1)T_{\mathrm{op}}. \tag{8.5}$$

Here T_c is source exposure and V_M bounds coherent memory space-time in qubit-time units. At fixed strengths and latency this layout takes $O(N^2)$ time and $O(N^3)$ memory space-time. Optimal native collision count does not imply optimal wall time for this layout. For weak θ, the resulting powers can be θ^{-8} for time and θ^{-12} for memory space-time, with accuracy logarithms.

8.2 A joint preparation bound and an exact finite factory

Suppose the actual L initial qubits are independent states τ_j. Put $x_j = 2\,\mathrm{Tr}(\tau_j^2) - 1$. Their global preparation error obeys

$$\frac{1}{2}\left\| \bigotimes_j \tau_j - (I/2)^{\otimes L} \right\|_1 \le \frac{1}{2}\sqrt{\prod_j (1 + x_j) - 1}. \tag{8.6}$$

For $Q = 2^L$, Cauchy–Schwarz gives $\|\rho - I/Q\|_1^2 \le Q \operatorname{Tr} \rho^2 - 1$, and product purity factors. Replacing an input-independent environment changes the resulting channel by at most this trace distance, uniformly in the payload/reference. Apply it once to the whole process, not repeatedly to correlated outputs.

For Bloch vectors of norm at most β, including the dummy so that $L = N + 1$, a sufficient condition for preparation error at most ε_p is

$$\beta^2 \le \frac{\log(1 + 4\varepsilon_p^2)}{L}. \tag{8.7}$$

This requires $\beta = O(\varepsilon_p/\sqrt{N})$. Local tracial marginals do not certify the product assumption. Correlated preparation needs a global bound or a sum of composable preparation-channel errors.

Exact tracial preparation need not mean literal infinite-temperature equilibration. Independent uniform Pauli randomization with inaccessible keys satisfies

$$\frac{1}{4} \sum_{P \in \{I,X,Y,Z\}} (P \otimes I)\rho(P^\dagger \otimes I) = I/2 \otimes \operatorname{Tr}_{\text{qubit}} \rho.$$

It removes initial correlations in the ideal gate model, costing two uniform classical bits and at most two Pauli pulses per cell, plus key generation and disposal. Every copy of the operating keys must be unavailable to receiver and future source controller. An accessible but unused key still supplies a nontracial conditional bath and changes the interface. Approximate twirls on correlated raw cells require composable channel-error bounds; they do not automatically justify the product formula.

8.3 Composable physical error and isolation

Let η_g bound each gate's half-diamond error uniformly with arbitrary entangled spectators. Let κ_S bound half the diamond norm of the unwanted source noise generator during collection. Let κ_M give a per-qubit bound whose sum controls the global memory generator. Direct composable interval errors can replace this generator model. Correlated noise needs a global bound; separate coherence times alone do not establish it.

Source isolation is a distinct requirement. Residual coupling to retained outputs supplies coherent feedback, outside the ideal interface. During every closed interval, including decoder operation while the source remains nearby, charge

$$\varepsilon_{\text{off}} := \int_{\text{closed}} \|H_{\text{off}}(t)\|_\infty \, \mathrm{d}t/\hbar \tag{8.8}$$

as a sufficient perturbation bound, or use a directly certified channel bound. It cannot be replaced by a source-only storage estimate.

Duhamel expansion, telescoping, and CPTP contraction give

$$\begin{aligned} d_\diamond(\widetilde{\mathcal{DC}}, \mathrm{id}) \le \tfrac{3}{4}(1 - \sin^4\theta)^N + \varepsilon_{\text{init}} + \sum_g \eta_g + \kappa_S T_c + \kappa_M V_M \\ + \varepsilon_{\text{off}} + \varepsilon_{\text{other}}. \end{aligned} \tag{8.9}$$

The last term covers otherwise uncounted leakage, drift, delay, key leakage, or cross-talk. Avoid double-counting errors already included elsewhere. This bound is reference uniform and assumes no independence of the emitted outputs.

One finite sufficient budget allocates $\varepsilon/2$ to the intrinsic first term and $\varepsilon/10$ each to preparation, gates, source storage, memory storage, and isolation, with no uncounted remaining errors. Thus

take

$$N \geq \frac{\log[(3/4)/(\varepsilon/2)]}{-\log(1-\sin^4\theta)}, \qquad \eta_g \leq \frac{\varepsilon}{10G}, \tag{8.10}$$

$$\beta^2 \leq \frac{\log(1+\varepsilon^2/25)}{N+1}, \qquad \kappa_S \leq \frac{\varepsilon}{10T_c}, \tag{8.11}$$

$$\kappa_M \leq \frac{\varepsilon}{10V_M}, \qquad \varepsilon_{\text{off}} \leq \frac{\varepsilon}{10}. \tag{8.12}$$

These conditions are conservative and sufficient. A protected logical implementation must add its encoding, factories, ancillas, gates, latency, and certified errors; none is supplied for free.

If actual emission is within ν of ideal emission in half-diamond distance, its optimal recovery error is at least the ideal lower bound minus ν. Its distance from its *own* replacer baseline is at most the ideal output bound plus 2ν. A fixed preparation bias cannot be discarded as N grows. For m independent physical modules, use the global intrinsic term in (6.6), $m(N+1)$ initialized cells, and the actual summed gate and space-time error budgets. Parallel operation reduces wall time but does not remove these summed physical error exposures.

8.4 Finite qualification and its exposure cost

Preparation qualification assumes stationary, reproducible, independent factories and trusted Pauli readout. For each of $L = N+1$ cell types, measure X, Y, Z on disposable preparations. If the empirical Bloch norm is at most $\beta/2$ and each mean has error at most $h \leq \beta/(2\sqrt{3})$, the true norm is at most β. Hoeffding's inequality and a union bound give

$$M_p \geq 24\beta^{-2}\log(6L/\alpha_p) \tag{8.13}$$

trials per axis and cell, or $3LM_p$ total. At $\beta = \Theta(\varepsilon_p/\sqrt{N})$, this is $O(N^2\varepsilon_p^{-2}\log(N/\alpha_p))$. Known preparation/readout systematics must fit inside the mean tolerance. These data do not certify independence, hidden-side-channel absence, or future stability. Calibration consumes no payload copies.

If exchange *form* is independently promised, prepare $|01\rangle$, apply the pulse, and measure $Z\otimes I$ and $X\otimes Y$ separately. Their means are $\cos(2\phi)$ and $\sin(2\phi)$, identifying angle modulo the irrelevant global phase period π. Angle precision h_ϕ costs $O(h_\phi^{-2}\log(K/\alpha_g))$ trials, also near zero because the second signal is linear there. Angle error $\delta\phi$ gives at most $|\delta\phi|$ half-diamond gate error. Inverse control needs qualification too. Scalar tests do not certify generic exchange form or absence of leakage and cross-talk.

A general certificate assumes a closed, stationary, context-independent two-qubit CPTP gate and trusted preparation/readout. For each distinct configuration, prepare its normalized Choi state using two input/reference Bell pairs, act on the two inputs, and measure all $L_P = 4^4-1 = 255$ nonidentity four-qubit Paulis separately. Each trial uses four qubits, two trusted Bell pairs, one tested gate, and four Pauli readouts. These references are paid calibration resources, not operating bath purifiers.

For input dimension $q = 4$, let $\widehat{r}_P$ estimate $r_P = \mathrm{Tr}(JP)$ with statistical width h and known systematic width s_P. A direct certificate is

$$d_\diamond(\mathcal{E},\mathcal{U}) \leq \frac{q}{2}\sqrt{\sum_{P\neq I}\left(|\widehat{r}_P - r_P^{\text{ideal}}| + h + s_P\right)^2}. \tag{8.14}$$

Pauli orthogonality and trace/Hilbert–Schmidt comparison imply $\|J_\mathcal{E} - J_\mathcal{U}\|_1 \leq \sqrt{\sum_P (\delta r_P)^2}$. Together with $\|\mathcal{E}-\mathcal{U}\|_\diamond \leq q\,\|J_\mathcal{E} - J_\mathcal{U}\|_1$ this proves the certificate without requiring a physical reconstructed map.

With K configurations, simultaneous statistical width h requires

$$M_g \geq 2h^{-2}\log(2L_PK/\alpha_g) \tag{8.15}$$

trials per Pauli and configuration. Take $h = O(\eta/(q\sqrt{L_P}))$ and control systematics similarly. Here $K \le 2N+1$: one source pulse, at most N receiver inverse locations, and N routing locations. With $G = O(N^2)$ and $\eta = \Theta(\varepsilon/G)$, the sufficient general qualification bill is

$$KL_PM_g = O\Big(N^5\varepsilon^{-2}\log(N/\alpha_g)\Big). \tag{8.16}$$

This is conservative; shared-parameter savings need independent justification. Low collision count does not hide the qualification cost.

Each gate trial takes actual Bell-preparation, pulse, readout, reset, and control time, multiplied by the displayed trial count and settings. Factory qualification takes $3LM_p$ actual preparation/readout/reset cycles. The ideal twirling factory additionally costs up to $2L$ Pauli pulses, $2L$ uniform random bits, and their inaccessible disposal. Trusted references and sensors cannot be obtained for free from the same unqualified devices. Source-edge tests precede payload loading or use a justified equivalent replica. On the joint confidence event, probability at least $1 - \alpha_p - \alpha_g$, the physical budget is reference uniform. Finite data qualify approximation under the stated model; they do not prove exact zero error, absence of all hidden interactions, or arbitrary future stability.

Global qualification for m modules. Without a shared-parameter reduction, the register construction has

$$G_{\mathrm{tot}} = m(N^2 + 5N - 2)/2, \quad K_{\mathrm{tot}} \le m(2N+1), \quad L_{\mathrm{tot}} = m(N+1).$$

Allocating a global gate error ε_g sets per-gate tolerance of order $\varepsilon_g/G_{\mathrm{tot}}$. Thus sufficient general gate qualification is

$$O\Big(m^3N^5\varepsilon_g^{-2}\log(mN/\alpha_g)\Big)$$

trials. A global product-preparation budget ε_p gives sufficient factory qualification

$$O\Big(m^2N^2\varepsilon_p^{-2}\log(mN/\alpha_p)\Big).$$

For parallel identical modules the elapsed operation time remains T_{op}, while memory space-time is at most $m(N+1)T_{\mathrm{op}}$. Serial execution also charges every initially present payload's waiting time before its own module begins. These explicit conservative bills may be reduced by justified common-parameter hardware, parallel test equipment, or improved certificates; none is silently assumed.

9 Executed evidence and resource evaluations

The accompanying program constructs the actual finite channels and inverse receiver. It does not replace proof by testing. In 48 dense qubit cases, $N = 1, \ldots, 6$, four base angles, and with noncommuting known source rotations in half the cases, the program retains the full 2^N-dimensional bath. It computes the operator Gram matrix and applies the inverse circuit to an independent mixed dummy after discarding the real source. The maximum Gram discrepancy is 9.9921×10^{-16}, and the maximum decoder-map discrepancy is 1.1103×10^{-15}.

Five additional dense cases use $d = 3, 4$, $N \le 3$, source rotations, and random mixed entangled input/reference states with reference dimension $d+1$. They evolve the joint state, trace out the actual source, append an independent dummy, and reverse the stored-output circuit. The largest entrywise discrepancy from (5.5) tensored with the reference identity is 5.563×10^{-17}.

Eighty generic two-qubit Hamiltonians test the single-round amplitude estimate $2a^2$; the largest observed ratio to that bound is 0.397592. Twelve selected feedback chains measure emitted qubits, keep their complete records, and choose subsequent interactions from preceding outcomes.

Their Bell-input trace distances satisfy the general bound. These are floating-point diagnostics of the implementation and conventions. They do not establish an all-Hamiltonian theorem or the full diamond norm by sampling. The unrestricted-decoder converse is analytic; no optimization of optimal decoders or hardware experiment was performed.

Modules	θ	Source forgetting	Every decoder necessary N	Reverse decoder sufficient N	Stored output qubits
1	0.05	1,727	581,933	691,949	691,949
1	0.10	432	36,552	43,462	43,462
1	0.20	108	2,329	2,770	2,770
1	0.40	27	157	186	186
1,000	0.10	—	105,970	112,948	112,948,000

Table 1: Qubit formula evaluations for target half-diamond error 0.01. Recovery counts N are per module; the last row targets the entire 1,000-qubit register and permits collective final decoding. Source forgetting targets a single source's distance from its replacer, so that column is omitted for the joint-register row. Single-module converse counts use the sharper centered bound (5.6). These are not simulations with the displayed numbers of bath qubits. Routing, precision, and memory costs are additional.

The table and both figures evaluate proved formulas. For example, $\theta = 0.1$ erases a source qubit to error 0.01 after 432 collisions, but every bath-only decoder needs at least 36,552, while the explicit receiver uses 43,462. An earlier coarse bound $c_d r_N/2$ gives the valid but weaker 36,485 count. The tighter value in Table 1 uses $c_d(1 - \sqrt{1 - r_N})$. For 1,000 independent modules at this angle, the explicit construction retains 112,948,000 qubits before accounting for protection overhead. This large cost is part of the result's practical interpretation.

These counts spend the entire 0.01 target on the ideal intrinsic error. Allocating only $\varepsilon/2$ to that error in (8.12) increases N before the physical preparation, gate, isolation, and storage costs are added.

The code, machine-readable records, exact resource formulas, and independent proof/selection notes accompany this manuscript. Large-resource examples are not claimed to have been physically executed, classically state-vector simulated, or supported by an implemented fault-tolerant memory.

10 Comparison with existing operations and accepted results

Quantum homogenization. Ziman and collaborators introduced sequential partial SWAP, source homogenization, correlation analysis, and a *quantum safe* inverse [1]. Their Section VII.B requires all $N + 1$ particles, identifies the original source, and reverses the interactions. The present receiver receives only the N emitted particles, substitutes an independent mixed dummy, and pays the exact residual r_N. The difference is physical access, not a new partial-SWAP gate or a new observation that correlations matter.

Yosifov and collaborators study homogenizer circuits and reservoir correctability using generalized Bény–Oreshkov conditions and a stipulated approximately forgetful complementary channel [2]. Their reviewed Section IV does not supply the finite tracial output-only $\sin^4\theta$ clock or the adaptive action bound here. We do not refute their conditional protection claim or equate their complement with the unavailable purifiers of our physical model.

Transpose recovery and physical inversion. Petz recovery, Barnum–Knill near-optimality, and Renes's sharper inequality are credited explicitly [3, 4]. Aw and collaborators give product preservation as a sufficient condition for physical inverse-unitary Petz implementation; Song and collaborators develop exact, approximate, and collisional implementation conditions [5, 6]. The

mixed-dummy inverse is an instance of that principle. Its generic existence is not our novelty claim. The calculation those principles alone do not provide is the acquired bath channel's complete information energy, including classical feedback, and its exact finite acquisition clock. Here the known reverse circuit also exposes its execution cost.

No-hiding, mixed dilations, and continuum limits. No-hiding refers to the complete pure dilation [7]. The physical mixed bath can be a strict part of that complement. Repeated-interaction limits have established mathematical machinery [8]. The present bound distinguishes the actual mixed output from a purified noise representation and charges the strength needed to keep source relaxation fixed. Violaris and Marletto study reservoir deterioration when homogenizers are reused for cyclic erasure [9]. That differs from recovering one input from all first-use outputs after source disposal; no conclusion about the economics of cyclic erasure is claimed.

Accepted Astra premises and nearby results. Corpus comparisons use revision `324acb589443002bf8d529cc15e160129c5e36d2` [10, 11, 12]. N1729 already gives a mixed classical-selector realization of a random-unitary source channel with input-independent emitted selector. It separately obtains a quartic finite-clock law for a *pure*-bath, two-collision driven-dephasing communication protocol. Neither mixed-implementation ambiguity nor the occurrence of a fourth power alone is a new consequence here.

N1733 proves that an extreme source channel's whole outgoing mixed bath is isometrically equivalent to its minimal complement tensored with the initial mixed bath, under explicit rank/access constraints. Its full-rank finite bath obstruction rules out a nonunitary extreme same-dimension source. The depolarizing source in the present regime is nonextreme; that positive result does not supply our receiver with a missing purifier.

The latest P1 collector supplies calibrated pure coherent ports for a many-body acquisition construction. Those are actual resources. The tracial once-through interface here does not possess them. Our result gives a quantitative boundary and attainable finite scale for a different interface, without altering the pure-port construction's conditions.

The increment and its limit. The local cancellation (3.7) becomes an all-output law through a contracting weighted norm and orthogonality of new and previously exported contributions. It survives receiver instruments and pathwise classical feedback. The partial-SWAP Gram identity supplies a matching operation, and existing recovery inequalities convert it to optimality bounds. Product-channel structure adds a joint-register cost, while equal exact source dynamics show why a reduced generator cannot replace an acquired-output specification.

The completed result certifies when a declared weak mixed-bath interface cannot yet carry a recoverable arbitrary state, and gives a receiver attaining the optimal scale with finite supplied resources. It does not identify an unknown source, certify every hidden coupling from finite data, recover from arbitrary thermal baths, or supply unlimited coherent memory. No exhaustive priority claim or civilization-scale consequence is established.

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